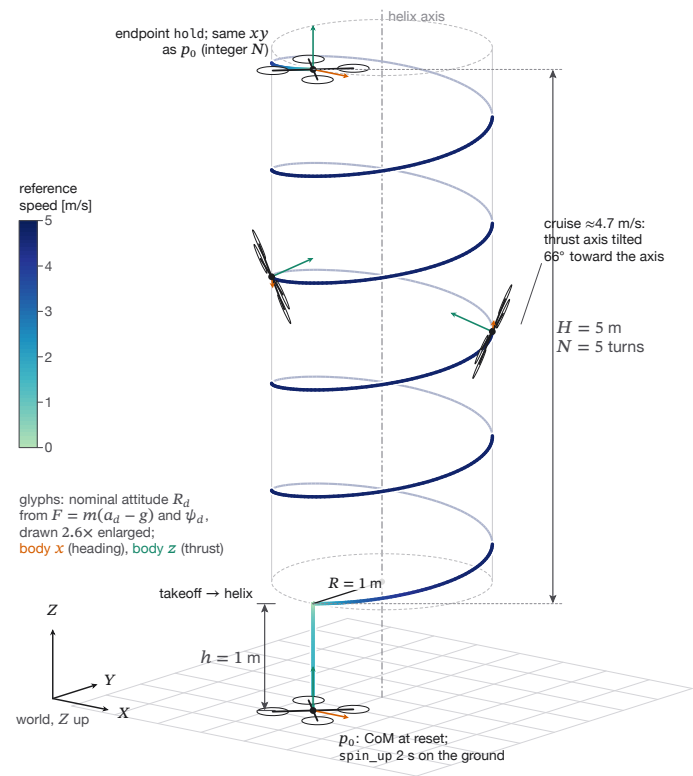
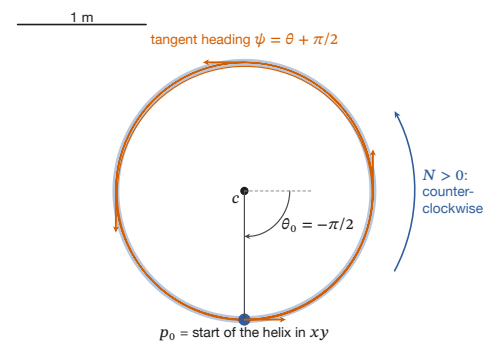


a **Planned mission** configs/helix.yaml, minimum-time plan



b **Top view** axis placed from the actual start



$c_{xy} = p_{0,xy} - R(\cos \theta_0, \sin \theta_0)$ puts p_0 on the circle, so the helix starts where the vertical takeoff ends (no lateral jump). Tangent heading at θ_0 equals the initial heading, so the takeoff needs no turn. Red: flown path.

c **Profiles** reference vs flown, synthetic plant

